Koopman Operator Theory in Drone Dynamics

Handwritten Notes Transcription & Formal Mathematical Derivations

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1. Drone Dynamics Framework

In drone dynamics, the general discrete-time state transition equation for a physical system is expressed as:

$$x_{k+1} = F(x_k, u_k)$$

Meaning of symbols: x_k is the state vector at time step k, u_k is the control input vector, and F is the true nonlinear state transition function.

For Linear Dynamic Mode Decomposition with Control (DMDc), the system evolution is approximated in linear state-space form as:

$$x_{k+1} = A\, x_k + B\, u_k$$

Meaning of symbols: A is the linear system state transition matrix and B is the control input matrix.

2. Mathematical Theory of the Koopman Operator

Consider an autonomous (uncontrolled) nonlinear dynamical system given by:

$$x_{k+1} = F(x_k)$$

Definition of an Observable g

Define a real-valued scalar or vector function g (an observable of the state x) such that:

$$g_{k+1} = g(F(x_k))$$

Explanation from notes: g applies a function to the next state of x_k given by $F(x_k)$.

The Koopman Operator K

The Koopman operator K is an infinite-dimensional linear operator acting on function space, defined by:

$$(K\,g)(x) = g(F(x))$$

Explanation from notes: The Koopman operator applied on g evolves x the same way g applied on F(x) does.

Lifting State Variables

Lifting is the process of applying the set of observables to the state x.

Let z be the vector of chosen observable functions:

$$z = [\, g_1(x) \;;\; g_2(x) \;;\; \ldots \,]_\{p \times 1\} \quad \Rightarrow \quad z = \psi(x)$$

Meaning of symbols: z is the p-dimensional lifted state vector, and ψ(x) is the lifting vector function that maps the n-dimensional state x into a p-dimensional observable space (where p > n).

3. Extended Dynamic Mode Decomposition (EDMD)

EDMD is a data-driven method used to estimate the finite-dimensional approximation of the Koopman operator K.

Step 1: Snapshot Collection

Collect sequential state transition snapshot pairs from x_1 to x_{m+1}:

Data Pairs: (x_1, x_2), (x_2, x_3), ..., (x_m, x_{m+1})

Step 2: Lifting Snapshot Data Matrices

Apply p observables to each of the snapshot states x_1 to x_{m+1}:

$$\Psi_X = [\, \psi(x_1) \;\; \psi(x_2) \;\; \ldots \;\; \psi(x_m) \,]_\{p \times m\}$$
$$\Psi_Y = [\, \psi(x_2) \;\; \psi(x_3) \;\; \ldots \;\; \psi(x_\{m+1\}) \,]_\{p \times m\}$$

Explanation & Dimensions: m snapshot states [x_1 ... x_m]_{1 × m}. Each snapshot is lifted using p observables (e.g., g_1(x) = x, g_2(x) = x^2, etc., forming a p × 1 lifted column vector).

Step 3: Estimating the Operator Matrix K

The linear transition between lifted snapshots satisfies:

$$\Psi_Y \approx K \cdot \Psi_X \quad \Rightarrow \quad K = \Psi_Y \cdot \Psi_X^{\dagger} \quad \textbf{(using Moore-Penrose Pseudoinverse)}$$

Meaning of symbols: K is the p × p estimated Koopman operator matrix, and Ψ_X† is the pseudoinverse of matrix Ψ_X.

The matrix K is calculated by solving the least-squares norm optimization problem:

$$K = \text{argmin}_K \, || \Psi_Y - K \cdot \Psi_X ||^2$$

4. Comparison of Dynamic Mode Decomposition Variants

The progression across modeling frameworks is summarized as follows:

1. Standard DMD: $x_{k+1} = A \cdot x_k$
2. DMD with Control (DMDc): $x_{k+1} = A\, x_k + B\, u_k$
3. Koopman on Autonomous DMD: $z_{k+1} \approx K \cdot z_k$
4. Koopman for Controlled Systems: $z_{k+1} \approx A\, z_k + B\, u_k$

Summary of Evolution: Progresses from state linear models to lifted state representations with external control inputs.

5. Koopman Framework for Controlled Systems

The Koopman operator K should be applied to the lifted state vector z.

Limitation of Ordinary EDMD: However, ordinary pseudoinverse estimation $K = \Psi_Y \cdot \Psi_X^{\dagger}$ does not include the B matrix (control inputs u_k). It only uses state data (X and Y lifted into Ψ_X and Ψ_Y).

Constructing the Combined Data Matrix Ω

To include control inputs, concatenate the lifted state snapshots Ψ_X with the control inputs U into an augmented matrix Ω:

$$\Omega = [\, \Psi_X \,;\, U \,]_{(p+n) \times m}$$

Meaning of symbols: Ω is the combined data matrix of size $(p + n) \times m$, where Ψ_X is $p \times m$ (lifted states) and U is $n \times m$ (control inputs u_k).

Solving for Joint Matrices [A B]

The evolution of the lifted state with control is formulated as:

$$\Psi_Y \approx [\, A \;\; B \,] \cdot [\, \Psi_X \,;\, U \,] = [\, A \;\; B \,] \cdot \Omega$$

Where [A B] is a joint matrix of size $p \times (p + n)$.

Solving for the joint matrices [A B] using the pseudoinverse Ω†:

> **$[\, A \quad B \,] = \Psi_Y \cdot \Omega^{\dagger}$**
>
> *Meaning of symbols: A is the p × p lifted state transition matrix, B is the p × n lifted control matrix, and Ω† is the pseudoinverse of Ω.*

6. State Reconstruction (Un-lifting)

To find the original physical state x back from the lifted vector z = ψ(x):

> **Extraction Rule: Simply take the rows corresponding to identity observables g_1(x) = x, g_2(x) = y, etc.**
>
> *Mathematical form: x = C z, where C is a state selection matrix [I_{n × n} 0] that projects the lifted vector z back to original state coordinates x.*

7. Core Advantages of Koopman Operator Theory

Why Koopman Operator Theory is used for complex drone control:

• **True System Dynamics:** x_{k+1} = F(x_k, u_k)

• **Standard Linearization Failure:** x_{k+1} = A x_k doesn't work directly because the drone dynamics are highly nonlinear.

• **Higher-Dimensional Lifting:** z_k = ψ(x_k) projects the state into a higher-dimensional space populated with different applied nonlinear functions.

• **Linear Predictability:** z_{k+1} ≈ K z_k enables linear predictions in the lifted space by relocating system nonlinearities into the coordinate transformation functions ψ(x).

8. Concrete Illustrative Example

The notes illustrate the observable transformation using a simple quadratic state transition example:

System Function: F(x) = x^2

Chosen Observables: g_1(x) = x, g_2(x) = x^2

Evaluating Observables on System Dynamics F(x)

Applying the observable functions to the next state transition F(x) yields:

> **g_1(F(x)) = g_1(x^2) = x^2**
> **g_2(F(x)) = g_2(x^2) = (x^2)^2 = x^4**

> *Explanation: Demonstrates how the Koopman operator maps observables across nonlinear state transitions in lifted functional space.*